PHYS 225 Practice Midterm

50 minutes | 100 points

1. Transporting a clock. Two synchronized clocks, A and B, are at rest in frame S at $x = 0$ and $x = L$, respectively. Both read t. Clock C moves in the $+x$ direction at constant speed v, with $0 < v < c$. It passes A at $t = 0$, when C also reads zero, and later passes B. Thus all three clocks read zero simultaneously in S. Let $\gamma = (1 - v^2/c^2)^{-1/2}$.

(a) (15 points) In frame S, find the time it takes C to reach B and the elapsed time recorded by C. Hence find the difference δ between B's and C's readings at their meeting, taking δ = (B's reading) - (C's reading).

(b) (15 points) Now work in C's inertial rest frame S', with $t' = 0$ at C's passage of A. Find B's reading at the event on B's worldline that is simultaneous in S' with that passage. Then use time dilation to calculate how much B's reading advances before it meets C. Explain quantitatively how your result agrees with part (a), even though B runs slowly in S'.

(c) (20 points) For fixed L and v/c much less than 1, use a Taylor expansion to find the first nonzero term in δ. Check its dimensions. Explain why δ can be made arbitrarily small even though the travel time becomes arbitrarily large.

2. Passing trains. On adjacent parallel tracks, train A has proper length L and moves east at $3c/5$, while train B has proper length $2L$ and moves west at $3c/5$, as measured in ground frame S. Event E_1 occurs when their fronts align; event E_2 occurs when their backs align. Treat the motion as one-dimensional. In all frames take east as positive, put the origins at E_1, and define every separation as E_2 minus E_1.

(a) (10 points) Using length contraction, find Δt and Δx between E_1 and E_2 in the ground frame.

(b) (20 points) Find the speed of one train relative to the other. Then use Lorentz transformations to find Δt_A and Δx_A in A's rest frame, and Δt_B and Δx_B in B's rest frame.

(c) (20 points) Calculate $\Delta s^2 = c^2(\Delta t)^2 - (\Delta x)^2$ in all three frames and classify the separation. Find the velocity (magnitude and direction) relative to the ground of a frame in which E_1 and E_2 occur at the same position. How much time elapses on a clock at rest at that position?

Adapted and extended from David Morin, Special Relativity for the Enthusiastic Beginner (2017): Problem 1.3 (Question 1) and Problem 2.6 (Question 2).